Author Manuscript

Author Manuscript

Author Manuscript

Author Manuscript

HHS Public Access

Author manuscript

Wien Klin Wochenschr. Author manuscript; available in PMC 2022 June 01.

Published in final edited form as:

Wien Klin Wochenschr. 2021 June ; 133(11-12): 586–593. doi:10.1007/s00508-020-01740-8.

Association between psoriasis, psoriatic arthritis and gastrointestinal disease:

An exploratory nationwide inpatient sample analysis

Ahmed Yousaf,
Department of Dermatology, West Virginia University, 64 Medical Center Drive, Morgantown, WV 26506, USA

Rahul Raiker,
Department of Dermatology, West Virginia University, 64 Medical Center Drive, Morgantown, WV 26506, USA

Stephen M. Davis,
Department of Health Policy, Management & Leadership, West Virginia University, Morgantown, WV, USA

Department of Emergency Medicine, West Virginia University, Morgantown, WV, USA

Swapna Gayam,
Department of Medicine, Section of Gastroenterology and Hepatology, West Virginia University, Morgantown, WV, USA

Zachary Zinn
Department of Dermatology, West Virginia University, 64 Medical Center Drive, Morgantown, WV 26506, USA

Summary

Background—Psoriasis is associated with cardiovascular disease, inflammatory bowel disease (IBD), metabolic syndrome, and psychiatric disease. Furthermore, psoriasis is associated with immune dysregulation and systemic inflammation.

Objective—To determine the association of psoriasis and psoriatic arthritis with IBD and the association of the combination of psoriasis or psoriatic arthritis with IBD and other gastrointestinal illnesses.

Methods—Discharge data from the 2000–2014 Nationwide Inpatient Sample, Healthcare Cost and Utilization Project (HCUP), which approximates a 20% stratified sample of all US hospitalizations, were analyzed. Multivariable logistic regression was used to examine the

zzinn@hsc.wvu.edu.

Conflict of interest A. Yousaf, R. Raiker, S.M. Davis, S. Gayam and Z. Zinn declare that they have no competing interests.

Compliance with ethical guidelines

Ethical standards Ethics approval: approved by the West Virginia University institutional review board, protocol # 2002899647.

Electronic supplementary material The online version of this article (https://doi.org/10.1007/s00508-020-01740-8) contains supplementary material, which is available to authorized users.

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

association between psoriasis and psoriatic arthritis with IBD and 23 gastrointestinal illnesses adjusting for sociodemographic characteristics.

Results—Psoriasis was associated with IBD (Crohn's disease adjusted odds ratio (aOR)=2.13, 95% confidence interval (CI) [2.0–2.3], $p<0.001$). When adjusting for sociodemographics and IBD, psoriasis was associated with 21 of 23 gastrointestinal diseases examined, most notably celiac disease, autoimmune hepatitis, and non-alcoholic fatty liver disease. Psoriatic arthritis was also associated with IBD (Crohn's disease, aOR=1.95, 95% CI [1.7–2.2], and ulcerative colitis, aOR= 2.66, 95% CI [2.4–2.9]).

Conclusion—Psoriasis and psoriatic arthritis inpatients have an associated increase in IBD and numerous other gastrointestinal illnesses.

Keywords

Crohn's disease; Ulcerative colitis; Liver disease; Celiac disease; Cirrhosis

Introduction

Psoriasis is a chronic, recurrent, immune-mediated inflammatory disorder with an estimated prevalence of 2–4% in western countries [1–3]. First presentation of psoriasis follows a bimodal distribution, peaking at ages 15–20 years and ages 55–60 years [4]. Psoriasis most commonly presents as well-defined, symmetric, erythematous plaques with silvery scale. Clinically benign skin from psoriatic patients may demonstrate epidermal thickening, increased DNA synthesis, parakeratosis, and infiltration of macrophages and lymphocytes in the dermis [5]. Although dermatologic manifestations are mostly visible, psoriasis is a systemic inflammatory disease.

Inflammatory bowel disease (IBD) is a chronic, recurrent, immune-mediated inflammatory disorder affecting 7% of the United States population [6]. It presents as two main forms: Crohn's disease (CD) and ulcerative colitis (UC), both resulting from a complex interaction of genetics, environment, and the gut microbiome. Population-based studies have shown an increasing incidence of IBD in the past century [7, 8].

Psoriasis and IBD have been shown to be closely connected diseases. Genome-wide association studies have revealed shared susceptibility loci between psoriasis and IBD, especially the 6p22, 16q, 1p31, and 5q33 loci, and interferon regulatory factor 5 [9, 10]. Furthermore, these conditions share pathogenetic mechanisms involving Th17 cells, regulatory T cells, and interleukins (IL) 17 and 23. Accordingly, methotrexate, TNF inhibitors, and IL-12/23 inhibitors can be used to treat psoriasis and IBD. Additional studies have shown common alterations in the gut microbiome [11, 12]. To date, risk associations between psoriasis and psoriatic arthritis with IBD have been reported [13–17], but not on a United States national scale. In this study, we quantify the prevalence and association of psoriasis and psoriatic arthritis with IBD and other gastrointestinal diseases in a nationwide inpatient sample.

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

Patients, material, methods

Data source

All data were extracted from the Nationwide Inpatient Sample (NIS), sponsored by the Healthcare Cost and Utilization Project (HCUP) [18], between 2000 and 2014. The dataset was intended for researchers to identify and analyze national trends in healthcare utilization, quality, and outcomes. The NIS contains an approximately 20% stratified, representative sample of all United States hospitalizations per year. Sample weights were created by NIS factoring sampling design of hospitals. These sample weights were needed to provide representative estimates of hospital discharges across the USA. All data were deidentified and no attempts were made to identify any of the individuals in the database. All parties with access to HCUP were compliant to the HCUP formal data use agreement. The NIS data concur with the National Hospital Discharge Survey, supporting data reliability [19].

Selection of psoriasis, psoriatic arthritis, and gastrointestinal illnesses

The databases were searched for a primary and/or secondary discharge diagnosis of psoriasis and psoriatic arthritis using the International Classification of Diseases, 9th revision, clinical modification (ICD-9-CM) code 696.1 and 696.0, respectively, both of which have been previously validated [20, 21]. The IBD was searched using ICD-9-CM codes previously validated in the NIS [22]. The primary diagnosis was defined in the NIS as the condition chiefly responsible for admission to hospital for care and per HCUP standards, is the first listed diagnosis. A maximum of 24 secondary diagnoses are allowed per discharge.

Gastrointestinal illnesses were selected based on those previously studied with psoriasis and other digestive diseases were selected a priori with unique ICD-9-CM codes for analysis [23]. A gastrointestinal disease was considered to be present with either a primary or secondary diagnosis. All comorbidities and their respective codes are presented in Supplemental Table 1.

Statistical analysis

Data were analyzed using SAS, Version 9.4 (SAS Institute, Cary, NC, USA). Independent variables were chosen to assess for sociodemographic disparities, as disparities in hospitalization rates have been observed in psoriasis and other dermatologic disorders, such as pemphigus and bullous pemphigoid [24–26]. Age, gender, race, and insurance status were included. Weighted frequencies of hospitalizations with either a primary or secondary ICD-9-CM code of psoriasis and psoriatic arthritis with and without IBD were determined for each independent variable. Summary statistics were generated for length of stay (LOS). Chi-square tests were employed to determine significant differences amongst all independent variables when comparing psoriasis with and without IBD and psoriatic arthritis with and without IBD. Significance was set to an alpha level of 0.05 a priori.

Multivariable logistic regression analyses were performed to identify the association of psoriasis and psoriatic arthritis with gastrointestinal illnesses. These analyses adjusted for age, gender, race, income, insurance status, obesity, smoking, and alcohol use when determining the association of psoriasis and psoriatic arthritis with and without IBD. All

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

other gastrointestinal illnesses were adjusted for IBD. Odds ratios (ORs) and 95% CIs were plotted on forest plots through Open Meta Analyst, an open-source software supported by Brown University and the Agency for Healthcare Research and Quality [27]. The standard error of the mean for each variable is reported as ±.

Cochran-Armitage temporal trend tests were performed for psoriasis and psoriatic arthritis with IBD at an alpha level of 0.05 and point-wise error bars representing 95% confidence intervals. The incidence of hospital cases is expressed as cases per 1000 hospitalizations. All analyses used SAS/STAT survey analysis procedures to account for the complex NIS sampling scheme and produce unbiased standard errors. *P*-values were not adjusted for multiple tests and should be interpreted as exploratory only.

Results

Patient hospitalization characteristics

The total weighted population captured by the NIS included 554,266,249 hospitalizations (116,134,755 unweighted) between 2000 and 2014. Among this sample, 21,355 hospitalizations had a primary or secondary diagnosis of psoriasis and IBD versus1,233,331 psoriasis hospitalizations without IBD. Psoriasis inpatients with vs. without IBD were much more likely to be younger (51± 0.3 years vs. 60± 0.1 years, $p<0.001$). Psoriasis inpatients with IBD were more likely to be female ($p<0.001$) and privately insured ($p<0.001$), while psoriasis inpatients without IBD were more likely to have pay insurance through Medicare ($p<0.001$). Psoriasis inpatients with IBD had significantly longer stays for inpatient hospitalizations as opposed to those without IBD (5.4± 0.10 days vs. 5.2± 0.02 days, $p=0.034$).

There were 5306 hospitalizations with a primary or secondary diagnosis of psoriatic arthritis and IBD versus 273,908 psoriatic arthritis hospitalizations without IBD. Psoriatic arthritis inpatients were mostly white, more often female, and between the ages of 40–79 years ($p<0.001$ each). Psoriatic arthritis hospitalizations with IBD were more likely to be female versus psoriatic arthritis hospitalizations without IBD ($p=0.034$). Psoriatic arthritis hospitalizations without IBD were disproportionately Hispanic ($p=0.004$). Like psoriasis hospitalizations with IBD, psoriatic arthritis hospitalizations with IBD had longer lengths of stay compared to those without IBD (5.4±0.2 days vs. 4.9±0.03 days, $p=0.004$). (Table 1).

Association of psoriasis with IBD and gastrointestinal illnesses

In multivariable regression models controlling for age, gender, race, income, insurance status, obesity, smoking, and alcohol use, psoriasis was significantly associated with IBD in both ulcerative colitis (adjusted odds ratio, aOR 2.13, 95% confidence interval, CI 2.0–2.3) and Crohn's disease (aOR 2.48, 95% CI 2.4–2.6). When also controlling for IBD, psoriasis was significantly associated with other gastrointestinal diseases, including celiac disease (aOR 2.62, 95% CI 2.3–2.9), gastroesophageal reflux disease (GERD) (aOR 1.53, 95% CI 1.5–1.6), esophagitis (aOR 1.05, 95% CI 1.0–1.1), peptic ulcers (aOR 1.22, 95% CI 1.1–1.3), autoimmune hepatitis (aOR 3.13, 95% CI 2.6–3.7), cholelithiasis (aOR 1.04, 95% CI 1.0–1.1), non-alcoholic fatty liver disease (aOR 2.22, 95% CI 2.1–2.3), gastritis and

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

duodenitis (aOR 1.04, 95% CI 1.0–1.1), cholangitis (aOR 1.21, 95% CI 1.1–1.4), irritable bowel syndrome (aOR 1.98, 95% CI 1.9–2.1), colon cancer (aOR 0.93, 95% CI 0.89–0.97), stomach cancer (aOR 0.73, 95% CI 0.65–0.82), esophageal cancer (aOR 0.87, 95% CI 0.77–0.99), pancreatic cancer (aOR 0.80, 95% CI 0.71–0.89), liver cancer (aOR 1.62, 95% CI 1.48–1.77), liver cirrhosis (aOR 2.95, 95% CI 2.85–3.06), noninfective gastroenteritis and colitis (aOR 1.07, 95% CI 1.03–1.12), intestinal infections (aOR 1.09, 95% CI 1.05–1.13), pernicious anemia (aOR 1.79, 95% CI 1.59–2.00), acute pancreatitis (aOR 1.09, 95% CI 1.05–1.13), and chronic pancreatitis (aOR 1.34, 95% CI 1.26–1.43) (Fig. 1).

Association of psoriatic arthritis with IBD and gastrointestinal illnesses

In multivariable regression models controlling for age, gender, race, income, insurance status, obesity, smoking, and alcohol use, psoriatic arthritis was significantly associated with IBD in both ulcerative colitis (aOR 1.95, 95% CI 1.7–2.2) and Crohn's disease (aOR 2.66, 95% CI 2.4–2.9). When also controlling for IBD, psoriatic arthritis was significantly associated with other gastrointestinal diseases, including celiac disease (aOR 2.63, 95% CI 2.1–3.2), GERD (aOR 1.91, 95% CI 1.8–2.0), esophagitis (aOR 1.33, 95% CI 1.2–1.5), peptic ulcers (aOR 1.63, 95% CI 1.5–1.7), autoimmune hepatitis (aOR 3.54, 95% CI 2.6–4.8), NAFLD (aOR 3.02, 95% CI 2.8–3.3), gastritis and duodenitis (aOR 1.32, 95% CI 1.2–1.4), irritable bowel syndrome (aOR 2.42, 95% CI 2.3–2.6), colon cancer (aOR 0.66, 95% CI 0.59–0.74), stomach cancer (aOR 0.63, 95% CI 0.47–0.85), esophageal cancer (aOR 0.54–0.98, 95% CI), liver cirrhosis (aOR 2.87, 95% CI 2.66–3.09), noninfective gastroenteritis and colitis (aOR 1.37, 95% CI 1.26–1.48), intestinal infections (aOR 1.39, 95% CI 1.29–1.48), pernicious anemia (aOR 2.48, 95% CI 2.01–3.05), acute pancreatitis (aOR 1.28, 95% CI 1.17–1.39), and chronic pancreatitis (aOR 1.58, 95% CI 1.37–1.82) (Fig. 2).

Trends of psoriasis and psoriatic arthritis with IBD

Among hospitalizations with psoriasis and psoriatic arthritis, rates of a concomitant diagnosis of IBD increased from 2000 to 2014 ($p<0.001$) (Fig. 3). In 2000, 11.7 psoriasis with IBD diagnoses per 1000 hospitalizations were made and in 2014, the number increased to 22.5 diagnoses per 1000 hospitalizations, representing a 92.3% increase. Likewise, in 2000, 12.2 psoriatic arthritis with IBD diagnoses per 1000 hospitalizations were made and in 2014, the number increased to 23 diagnoses per 1000 hospitalizations showing an 88.5% rise across a 15-year period.

Discussion

In this study, we show that weighted frequencies of sex, age, race, and insurance status were significantly different amongst psoriasis inpatients with vs. without IBD and psoriatic arthritis inpatients with vs. without IBD. Psoriasis inpatients with vs. without IBD were much more likely to be younger, female, and privately insured. Psoriasis inpatients without IBD were more likely to be insured with Medicare. Psoriatic arthritis inpatients with vs. without IBD were much more likely to be white, female, and between the ages of 40–79 years. Both psoriasis and psoriatic arthritis with vs. without IBD had longer lengths of inpatient hospitalization, likely due to increased complexity of comorbid health disorders.

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

The present study highlights the association of psoriasis and psoriatic arthritis with gastrointestinal diseases, particularly IBD, even after controlling for demographic variables. The strongest associations in psoriasis inpatients were IBD, celiac disease, autoimmune hepatitis, NAFLD, and liver cirrhosis. Esophageal, stomach, colon, and pancreatic cancer all had lower odds, while liver cancer had higher odds in psoriasis inpatients. Psoriatic arthritis inpatients shared the associations with liver diseases, as well as a particularly strong association with IBS and pernicious anemia.

Although psoriasis patients have been shown to smoke and drink alcohol more than other dermatology patients, both lifestyle factors were controlled for in multivariable regression models [28, 29]. Nonetheless, psoriasis and psoriatic arthritis showed increased odds of developing liver diseases, including liver cancer, autoimmune hepatitis, NAFLD, and liver cirrhosis. The effects of psoriasis and psoriatic arthritis on the liver remain unclear. In the current study, past use of methotrexate, a hepatotoxic drug used in psoriasis, could not be obtained. Previous studies have shown increased risk of liver disease in psoriasis and psoriatic arthritis patients [30–32]. It has been documented that autoimmune hepatitis is a complication of psoriatic anti-tumor necrosis factor α treatment due to the production of autoantibodies. This study stresses the importance of careful monitoring of liver enzymes in psoriasis and psoriatic arthritis patients, especially when treated with hepatotoxic drugs. Further studies are needed to investigate why there is a negative association with psoriasis and psoriatic arthritis with other gastrointestinal cancers as previous reports suggest increased risk [31, 33–35].

Previous studies have explored the association of psoriasis and psoriatic arthritis with IBD. Possible theories explaining this association include genetic abnormalities, immune dysfunction, and alteration of gut microbiota [23]. Chromosomal locus 6p21 is the most studied genetic region, encoding major histocompatibility complex-related genes [36]. Both psoriasis and IBD share susceptibility loci on chromosome 6p21, with psoriasis linked to *PSORS1* and IBD to *IBD3* [9]. Moreover, immune dysfunction affecting interleukin 23, 17, and 12 have been found in the pathogenesis of psoriasis and IBD [37–40]. Psoriasis patients have been shown to have decreased gut microbiome diversity, similar to IBD patients [11, 41].

To our knowledge, prior studies have not examined the trend of psoriasis and psoriatic arthritis with IBD in hospitalized patients. This nationwide analysis of hospital discharges demonstrates that psoriasis inpatients with IBD and psoriatic arthritis inpatients with IBD have experienced a considerable absolute rise in concomitant diagnoses. In part, this may be due to rising prevalence of psoriasis and IBD overall [42, 43]. Given that psoriasis and IBD are more prevalent in older age groups, it is likely that the absolute rise in psoriasis with IBD will continue to increase as the average US lifespan is forecasted to increase.

Strengths of the study include the size and diversity of the patient population in sex, age, race, and insurance status in a nationally representative sample. Moreover, the study used previously validated ICD-9-CM codes known to have high positive predictive value for psoriasis, psoriatic arthritis, and IBD.

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

Limitations of the study include exclusively reviewing an inpatient population, suggesting that Berkson's bias may be present, overestimating severe disease phenotypes since these cases are more prone to hospitalization. This warrants further studies examining the observed trends in an outpatient setting. Furthermore, medication use was not accessible for review. Consequently, the effect of psoriatic therapeutics, particularly methotrexate and biologics, may confound the results. Also, since the data set is based on discharge diagnoses, an individual patient with multiple admissions could have been counted multiple times, introducing bias. Lastly, outside the analysis of psoriasis and psoriatic arthritis with IBD, this was a cross-sectional study that did not explore trends of psoriasis and psoriatic arthritis with other gastrointestinal diseases. Therefore, it remains unknown if these associations have been rising or decreasing over the 15-year period studied.

In conclusion, psoriasis and psoriatic arthritis are associated with IBD and a number of gastrointestinal diseases. Significant gender, age, racial, and healthcare disparities exist in hospitalized psoriasis and psoriatic arthritis inpatients with vs. without IBD. The incidence of hospitalizations for psoriasis and psoriatic arthritis with IBD significantly increased between 2000 and 2014. Further studies are needed to confirm these findings and uncover the mechanisms tying psoriasis and psoriatic arthritis to gastrointestinal diseases.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

References

1. Stern RS, Nijsten T, Feldman SR, Margolis DJ, Rolstad T, (eds.). Psoriasis is common, carries a substantial burden even when not extensive, and is associated with widespread treatment dissatisfaction. Journal of Investigative Dermatology Symposium Proceedings; 2004 3;9(2):136–9. 10.1046/j.1087-0024.2003.09102.x.
2. Gelfand JM, Weinstein R, Porter SB, Neimann AL, Berlin JA, Margolis DJ. Prevalence and treatment of psoriasis in the United Kingdom: a population-based study. Arch Dermatol. 2005;141(12):1537–41. [PubMed: 16365254]
3. Kurd SK, Gelfand JM. The prevalence of previously diagnosed and undiagnosed psoriasis in US adults: results from NHANES 2003–2004. J Am Acad Dermatol. 2009;60(2):218–24. [PubMed: 19022533]
4. Smith A, Kassab J, Payne R, Beer W. Bimodality in age of onset of psoriasis, in both patients and their relatives. Dermatology.1993;186(3):181–6. [PubMed: 8453143]
5. Farber EM, Nall L, Strefling A. Psoriasis:a disease of the total skin. J Am Acad Dermatol. 1985;12(1):150–6. [PubMed: 3884677]
6. Friedman S, Rubin PH, Bodian C, Goldstein E, Harpaz N, Present DH. Screening and surveillance colonoscopy in chronic Crohn's colitis. Gastroenterology. 2001;120(4):820–6. [PubMed: 11231935]
7. Loftus EV Jr, Silverstein MD, Sandborn WJ, Tremaine WJ, Harmsen WS, Zinsmeister AR. Crohn's disease in Olmsted County,Minnesota,1940–1993:incidence, prevalence, and survival. Gastroenterology. 1998;114(6):1161–8. [PubMed: 9609752]
8. Loftus E, Silverstein M, Sandborn W, Tremaine W, Harmsen W, Zinsmeister A. Ulcerative colitis in Olmsted County, Minnesota, 1940–1993: incidence, prevalence, and survival. Gut. 2000;46(3):336–43. [PubMed: 10673294]
9. Skroza N, Proietti I, Pampena R, La Viola G, Bernardini N, Nicolucci F, et al. Correlations between psoriasis and inflammatory bowel diseases. Biomed Res Int. 2013; 10.1155/2013/983902.

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

10. Sánchez FO, Reddy MPL, Sakuraba K, Ståhle M, Alarcón-Riquelme ME. IFN-regulatory factor 5 gene variants interact with the class I MHC locus in the Swedish psoriasis population. J Investig Dermatol. 2008;128(7):1704–9. [PubMed: 18200047]
11. Eppinga H, Sperna Weiland CJ, Thio HB, van der Woude CJ, Nijsten TE, Peppelenbosch MP, et al. Similar depletion of protective faecalibacterium prausnitzii in psoriasis and inflammatory bowel disease, but not in hHidradenitiss Suppurativa. J Crohns Colitis. 2016;10(9):1067–75.10.1093/ecco-jcc/jjw070. [PubMed: 26971052]
12. Cottone M, Sapienza C, Macaluso FS, Cannizzaro M. Psoriasis and inflammatory bowel disease. Dig Dis. 2019;37(6):451–7.10.1159/000500116. [PubMed: 31079092]
13. Vavricka SR, Brun L, Ballabeni P, Pittet V, Vavricka BMP, Zeitz J, et al. Frequency and risk factors for extraintestinal manifestations in the Swiss inflammatory bowel disease cohort. Am J Gastroenterol. 2011;106(1):110–9. [PubMed: 20808297]
14. Cohen A, Dreiher J, Birkenfeld S. Psoriasis associated with ulcerative colitis and Crohn's disease. J Eur Acad Dermatol Venereol. 2009;23(5):561–5. 10.1111/j.1468-3083.2008.03031.x. [PubMed: 19207663]
15. Makredes M, Robinson D Jr, Bala M, Kimball AB. The burden of autoimmune disease: a comparison of prevalence ratios in patients with psoriatic arthritis and psoriasis. J Am Acad Dermatol. 2009;61(3):405–10. [PubMed: 19700012]
16. Bernstein CN, Wajda A, Blanchard JF. The clustering of other chronic inflammatory diseases in inflammatory bowel disease: a population-based study. Gastroenterology. 2005;129(3):827–36. [PubMed: 16143122]
17. Persson PG, Leijonmarck CE, Bernell O, Hellers G, Ahlbom A. Risk indicators for inflammatory bowel disease. Int J Epidemiol. 1993;22(2):268–72. [PubMed: 8505183]
18. Agency for Healthcare Research and Quality. HCUP databases. Healthcare cost and utilization project (HCUP). 2019. www.hcup-us.ahrq.gov/nisoverview.jsp. Accessed 1 March 2020.
19. Whalen D, Houchens R, Elixha-user A. 2002 HCUP nationwide inpatient sample (NIS) comparison report. HCUP method series report# 2007–03. Rockville: US Agency for Healthcare Research and Quality; 2007.
20. Icen M, Crowson CS, McEvoy MT, Gabriel SE, Maradit Kremers H. Potential misclassification of patients with psoriasis in electronic databases. J Am Acad Dermatol. 2008;59(6):981–5. 10.1016/j.jaad.2008.08.034. [PubMed: 18835060]
21. Asgari MM, Wu JJ, Gelfand JM, Salman C, Curtis JR, Harrold LR, et al. Validity of diagnostic codes and prevalence of psoriasis and psoriatic arthritis in a managed care population, 1996–2009. Pharmacoepidemiol Drug Saf. 2013;22(8):842–9. 10.1002/pds.3447. [PubMed: 23637091]
22. Peery AF, Dellon ES, Lund J, Crockett SD, McGowan CE, Bulsiewicz WJ, et al. Burden of gastrointestinal disease in the United States: 2012 update. Gastroenterology. 2012;143(5):1179–1187.e3. 10.1053/j.gastro.2012.08.002. [PubMed: 22885331]
23. Fu Y, Lee CH, Chi CC. Association of psoriasis with inflammatory bowel disease: a systematic review and meta-analysis. JAMA Dermatol. 2018;154(12):1417–23. 10.1001/jamadermatol.2018.3631. [PubMed: 30422277]
24. Ren Z, Hsu DY, Brieva J, Silverberg NB, Langan SM, Silverberg JI. Hospitalization, inpatient burden and comorbidities associated with bullous pemphigoid in the U.S.A. Br J Dermatol. 2017;176(1):87–99. 10.1111/bjd.14821. [PubMed: 27343837]
25. Hsu D, Brieva J, Silverberg JI. Costs of care for hospitalization for pemphigus in the United States. JAMA Dermatol. 2016;152(6):645–54. 10.1001/jamadermatol.2015.5240. [PubMed: 26865293]
26. Hsu DY, Gordon K, Silverberg JI. Serious infections in hospitalized patients with psoriasis in the United States. J Am Acad Dermatol. 2016;75(2):287–96. 10.1016/j.jaad.2016.04.005. [PubMed: 27324314]
27. Wallace BC, Dahabreh IJ, Trikalinos TA, Lau J, Trow P, Schmid CH. Closing the gap between methodologists and end-users: R as a computational back-end. J Stat Softw. 2012;49(5):1–15.
28. Poikolainen K, Reunala T, Karvonen J. Smoking, alcohol and life events related to psoriasis among women. Br J Dermatol. 1994;130(4):473–7. [PubMed: 8186112]
29. Poikolainen K, Reunala T, Karvonen J, Lauharanta J, Kärkkäinen P. Alcohol intake: a risk factor for psoriasis in young and middle aged men? BMJ. 1990;300(6727):780–3. [PubMed: 1969757]

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

30. Chen Y-J, Wu C-Y, Chen T-J, Shen J-L, Chu S-Y, Wang C-B, et al. The risk of cancer in patients with psoriasis: a population-based cohort study in Taiwan. J Am Acad Dermatol. 2011;65(1):84–91. [PubMed: 21458106]
31. Boffetta P, Gridley G, Lindelöf B. Cancer risk in a population-based cohort of patients hospitalized for psoriasis in Sweden. J Investig Dermatol. 2001;117(6):1531–7. [PubMed: 11886519]
32. Miele L, Vallone S, Cefalo C, La Torre G, Di Stasi C, Vecchio FM, et al. Prevalence, characteristics and severity of non-alcoholic fatty liver disease in patients with chronic plaque psoriasis. J Hepatol. 2009;51(4):778–86. [PubMed: 19664838]
33. Frentz G, Olsen J. Malignant tumours and psoriasis: a follow-up study. Br J Dermatol. 1999;140(2):237–42. [PubMed: 10233215]
34. Hannuksela-Svahn A, Pukkala E, Läärä E, Poikolainen K, Karvonen J. Psoriasis, its treatment, and cancer in a cohort of Finnish patients. J Investig Dermatol. 2000;114(3):587–90. [PubMed: 10692122]
35. Stern RS, Lange R. Cardiovascular disease, cancer, and cause of death in patients with psoriasis: 10 years prospective experience in a cohort of 1,380 patients. J Investig Dermatol. 1988;91(3):197–201. 10.1111/1523-1747.ep12464847. [PubMed: 3411140]
36. Ahmad T, Marshall SE, Jewell D. Genetics of inflammatory bowel disease: the role of the HLA complex. World J Gastroenterol. 2006;12(23):3628. [PubMed: 16773677]
37. Capon F, Di Meglio P, Szaub J, Prescott NJ, Dunster C, Baumber L, et al. Sequence variants in the genes for the interleukin-23 receptor (IL23R) and its ligand (IL12B) confer protection against psoriasis. Hum Genet. 2007;122(2):201–6. [PubMed: 17587057]
38. Maddur MS, Miossec P, Kaveri SV, Bayry J. Th17 cells:biology, pathogenesis of autoimmune and inflammatory diseases, and therapeutic strategies. Am J Pathol. 2012;181(1):8–18. 10.1016/j.ajpath.2012.03.044. [PubMed: 22640807]
39. Chiricozzi A, Krueger JG. IL-17 targeted therapies for psoriasis. Expert Opin Investig Drugs. 2013;22(8):993–1005.
40. Fujino S, Andoh A, Bamba S, Ogawa A, Hata K, Araki Y, et al. Increased expression of interleukin 17 in inflammatory bowel disease. Gut. 2003;52(1):65–70. [PubMed: 12477762]
41. Scher JU, Ubeda C, Artacho A, Attur M, Isaac S, Reddy SM, et al. Decreased bacterial diversity characterizes the altered gut microbiota in patients with psoriatic arthritis, resembling dysbiosis in inflammatory bowel disease. Arthritis Rheumatol.2015;67(1):128–39. [PubMed: 25319745]
42. Icen M, Crowson CS, McEvoy MT, Dann FJ, Gabriel SE, Maradit Kremers H. Trends in incidence of adult-onset psoriasis over three decades: a population-based study. J Am Acad Dermatol. 2009;60(3):394–401. 10.1016/j.jaad.2008.10.062. [PubMed: 19231638]
43. Kappelman MD, Moore KR, Allen JK, Cook SF. Recenttrends in the prevalence of Crohn's disease and ulcerative colitis in a commercially insured US population. Dig Dis Sci. 2013;58(2):519–25. [PubMed: 22926499]

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

Gastrointestinal Disease	Psoriasis	
	aOR [95% CI]	p-value
Ulcerative Colitis	**2.13 [2.0-2.3]**	**<0.001**
Crohn's Disease	**2.48 [2.4-2.6]**	**<0.001**
Celiac Disease	**2.62 [2.3-2.9]**	**<0.001**
GERD	**1.53 [1.5-1.6]**	**<0.001**
Esophagitis	**1.05 [1.0-1.1]**	**0.049**
Peptic Ulcers	**1.22 [1.1-1.3]**	**<0.001**
Autoimmune Hepatitis	**3.13 [2.6-3.7]**	**<0.001**
Cholecystitis	0.94 [0.9-1.1]	0.092
Cholelithiasis	**1.04 [1.0-1.1]**	**0.044**
NAFLD	**2.22 [2.1-2.3]**	**<0.001**
Gastritis-Duodenitis	**1.04 [1.0-1.1]**	**0.034**
Cholangitis	**1.21 [1.1-1.4]**	**<0.001**
IBS	**1.98 [1.9-2.1]**	**<0.001**
Colon Cancer	**0.93 [0.89-0.97]**	**0.012**
Stomach Cancer	**0.73 [0.65-0.82]**	**<0.001**
Esophageal Cancer	**0.87 [0.77-0.99]**	**0.041**
Pancreatic Cancer	**0.80 [0.71-0.89]**	**<0.001**
Liver Cancer	**1.62 [1.48-1.77]**	**<0.001**
Liver Cirrhosis	**2.95 [2.85-3.06]**	**<0.001**
Intestinal Fistula	1.06 [0.89-1.24]	0.535
Noninfective Gastroenteritis and Colitis	**1.07 [1.03-1.12]**	**<0.001**
Intestinal Infections	**1.09 [1.05-1.13]**	**<0.001**
Pernicious Anemia	**1.79 [1.59-2.00]**	**<0.001**
Acute Pancreatitis	**1.09 [1.05-1.13]**	**<0.001**
Chronic Pancreatitis	**1.34 [1.26-1.43]**	**<0.001**

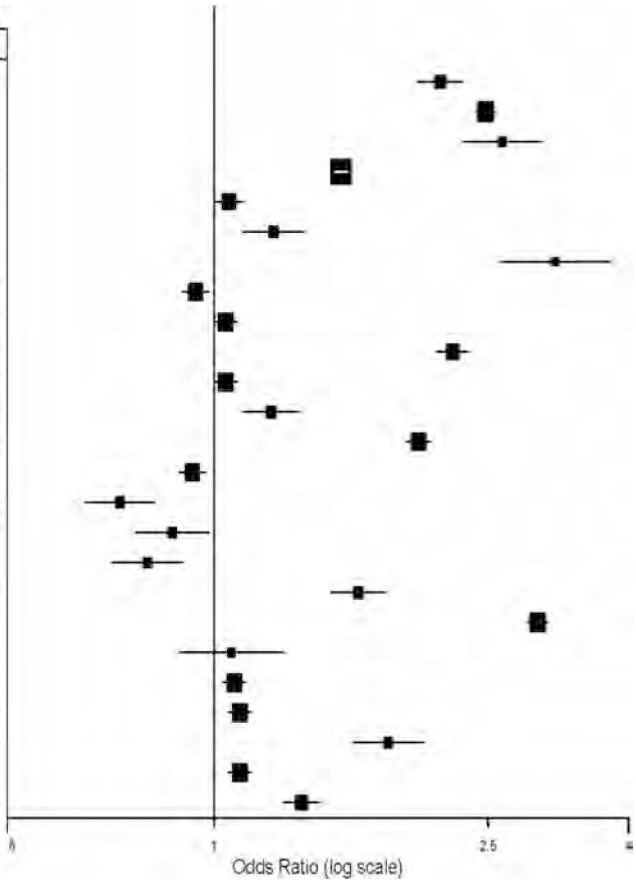


Fig. 1.

Association between psoriasis and gastrointestinal comorbidities. Significant adjusted odds ratios, 95% confidence intervals, and *p*-values in bold. *aOR* adjusted odds ratio, *GERD* gastroesophageal reflux disease, *NAFLD* non-alcoholic fatty liver disease, *IBS* irritable bowel syndrome

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

Gastrointestinal Disease	Psoriatic Arthritis	
	aOR [95% CI]	p-value
Ulcerative Colitis	**1.95 [1.7-2.2]**	**<0.001**
Crohn's Disease	**2.66 [2.4-2.9]**	**<0.001**
Celiac Disease	**2.63 [2.1-3.2]**	**<0.001**
GERD	**1.91 [1.8-2.0]**	**<0.001**
Esophagitis	**1.33 [1.2-1.5]**	**<0.001**
Peptic Ulcers	**1.63 [1.5-1.7]**	**<0.001**
Autoimmune Hepatitis	**3.54 [2.6-4.8]**	**<0.001**
Cholecystitis	1.08 [0.9-1.3]	0.299
Cholelithiasis	0.97 [0.9-1.1]	0.722
NAFLD	**3.02 [2.8-3.3]**	**<0.001**
Gastritis-Duodenitis	**1.32 [1.2-1.4]**	**<0.001**
Cholangitis	1.1 [0.9-1.4]	0.448
IBS	**2.42 [2.3-2.6]**	**<0.001**
Colon Cancer	**0.66 [0.59-0.74]**	**<0.001**
Stomach Cancer	**0.63 [0.47-0.85]**	**0.003**
Esophageal Cancer	**0.73 [0.54-0.98]**	**0.030**
Pancreatic Cancer	0.90 [0.73-1.12]	0.346
Liver Cancer	0.99 [0.78-1.26]	0.933
Liver Cirrhosis	**2.87 [2.66-3.09]**	**<0.001**
Intestinal Fistula	0.97 [0.70-1.35]	0.870
Noninfective Gastroenteritis and Colitis	**1.37 [1.26-1.48]**	**<0.001**
Intestinal Infections	**1.39 [1.29-1.48]**	**<0.001**
Pernicious Anemia	**2.48 [2.01-3.05]**	**<0.001**
Acute Pancreatitis	**1.28 [1.17-1.39]**	**<0.001**
Chronic Pancreatitis	**1.58 [1.37-1.82]**	**<0.001**

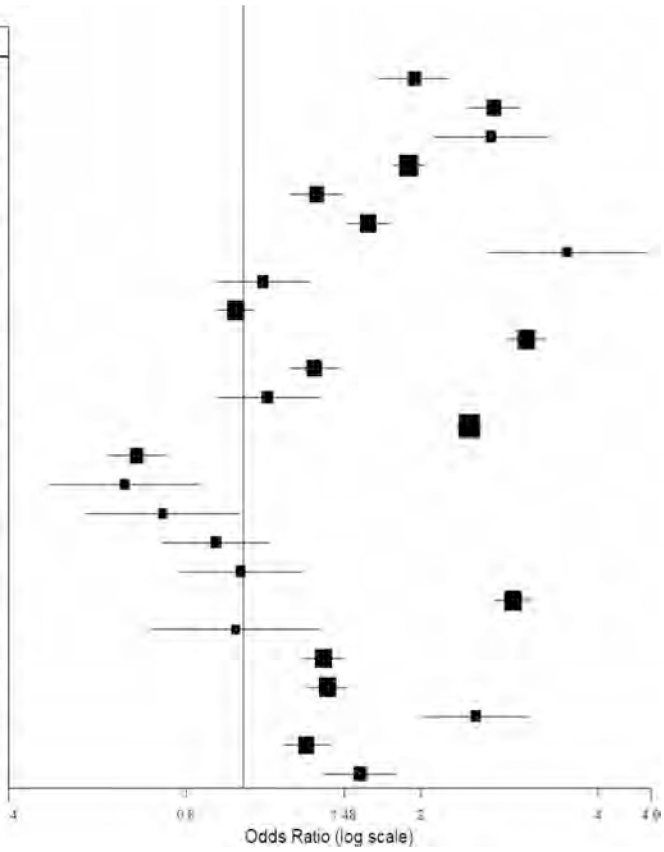


Fig. 2.

Association between psoriatic arthritis and gastrointestinal comorbidities. Significant adjusted odds ratios, 95% confidence intervals, and *p*-values in bold. *aOR* adjusted odds ratio, *GERD* gastroesophageal reflux disease, *NAFLD* non-alcoholic fatty liver disease, *IBS* irritable bowel syndrome

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

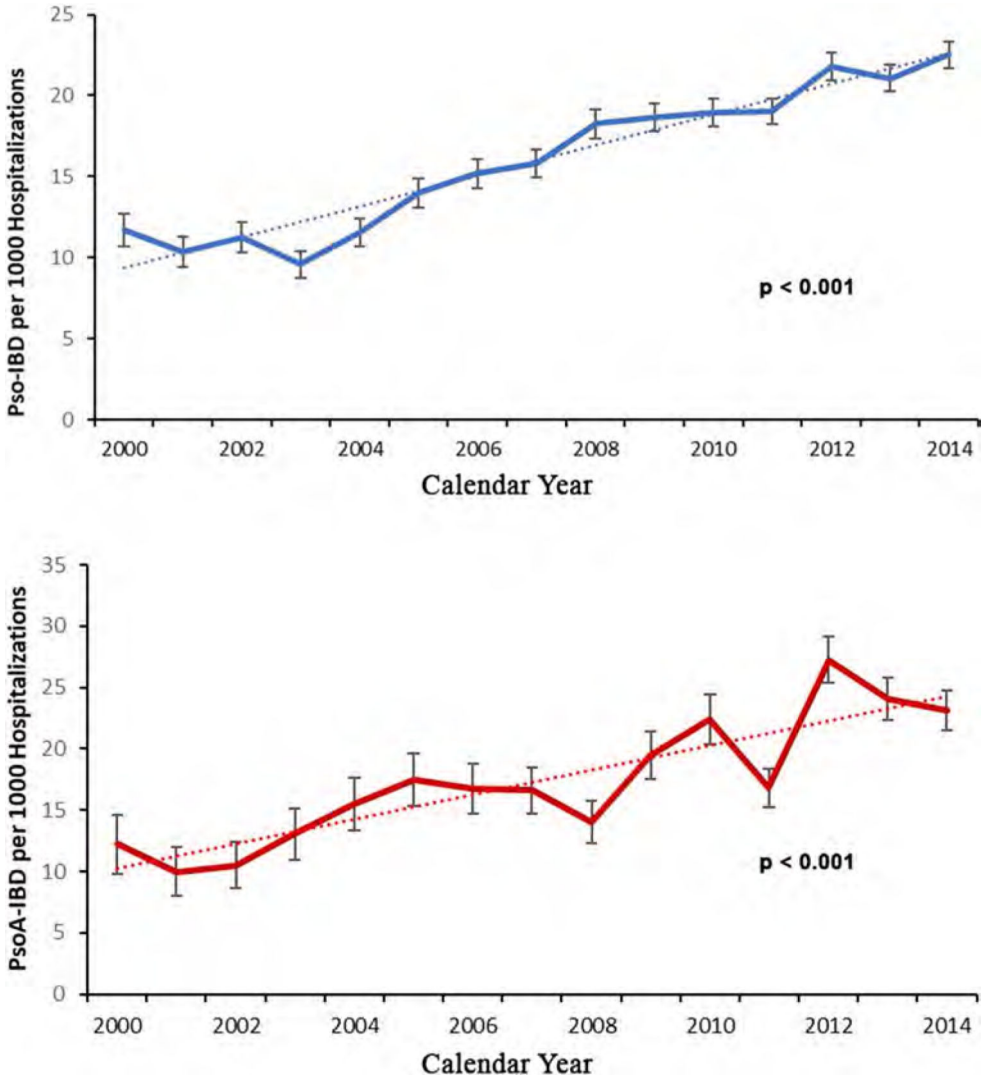


Fig. 3.

Incidence of hospitalization for psoriasis and psoriatic arthritis with IBD. Incidence of hospitalization for both psoriasis and psoriatic arthritis with IBD (blue and red, respectively) per 1000 hospitalizations are rising. Estimates and 95% confidence intervals are presented

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

Table 1

Patient hospitalization characteristics

Variable	Psoriasis			Psoriatic arthritis		
	w/IBD	w/o IBD	*P*-value	w/IBD	w/o IBD	*P*-value
Sex	–	–	<0.001*	–	–	0.034*
Female	11,750 (55.0)	573,519 (47.0)	–	3157 (60.0)	152,160 (56.0)	–
Male	9605 (45.0)	638,167 (53.0)	–	2149 (40.0)	121,666 (44.0)	–
Age, years	–	–	<0.001*	–	–	<0.001*
0–17	420 (2.0)	8646 (0.7)	–	47 (0.9)	931 (0.3)	–
18–39	5786 (27.0)	133,749 (11.3)	–	818 (15.0)	24,026 (8.8)	–
40–59	7889 (37.0)	420,650 (35.4)	–	2356 (44.0)	107,055 (39.0)	–
60–79	6015 (28.0)	493,187 (41.5)	–	1854 (35.0)	118,003 (43.0)	–
>80	1244 (6.0)	132,191 (11.1)	–	231 (4.4)	23,893 (8.7)	–
Race	–	–	<0.001*	–	–	0.004*
White	16,083 (88.0)	823,323 (81.0)	–	4308 (92.6)	203,593 (88.4)	–
Black	902 (4.9)	67,990 (6.7)	–	123 (2.6)	7153 (3.1)	–
Hispanic	634 (3.5)	76,453 (7.5)	–	96 (2.1)	10,546 (4.6)	–
Asian	211 (1.1)	21,620 (2.1)	–	24 (0.5)	3290 (1.4)	–
Native American	49 (0.2)	5249 (0.5)	–	24 (0.5)	1017 (0.5)	–
Other	420 (2.3)	25,262 (2.5)	–	78 (1.7)	4677 (2.0)	–
Insurance	–	–	<0.001*	–	–	<0.001*
Medicare	7742 (36.3)	597,425 (49.4)	–	2205 (41.6)	131,768 (48.2)	–
Medicaid	2174 (10.2)	138,380 (11.4)	–	355 (6.7)	19,076 (7.0)	–
Private	9782 (45.8)	370,222 (30.6)	–	2491 (46.9)	108,819 (39.8)	–
Self-pay	873 (4.1)	57,977 (4.8)	–	87 (1.6)	6371 (2.3)	–
No charge	97 (0.5)	6363 (0.5)	–	*a*	667 (0.2)	–
Other	671 (3.1)	39,311 (3.3)	–	169 (3.2)	6869 (2.5)	–
Length of stay, days (SE)	5.4 (± 0.10)	5.2 (± 0.02)	0.034*	5.4 (± 0.2)	4.9 (± 0.03)	0.004*

Values are presented as *n* (%) unless specified. Percent applies to column.

Wien Klin Wochenschr. Author manuscript; available in PMC 2022 June 01.

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

IBD inflammatory bowel disease

*Significant at level $P = 0.05$

[a]Per the Nationwide Inpatient Sample data use agreement, cell sizes ≤10 should be avoided in publication. Missing values were encountered for age in 23,264, race in 194,845, and insurance in 2024 patients with psoriasis. Missing values were encountered for sex in 82, race in 44,203, and insurance in 256 patients with psoriatic arthritis. There were no significant differences between missing values among patients with or without IBD

Wien Klin Wochenschr. Author manuscript; available in PMC 2022 June 01.